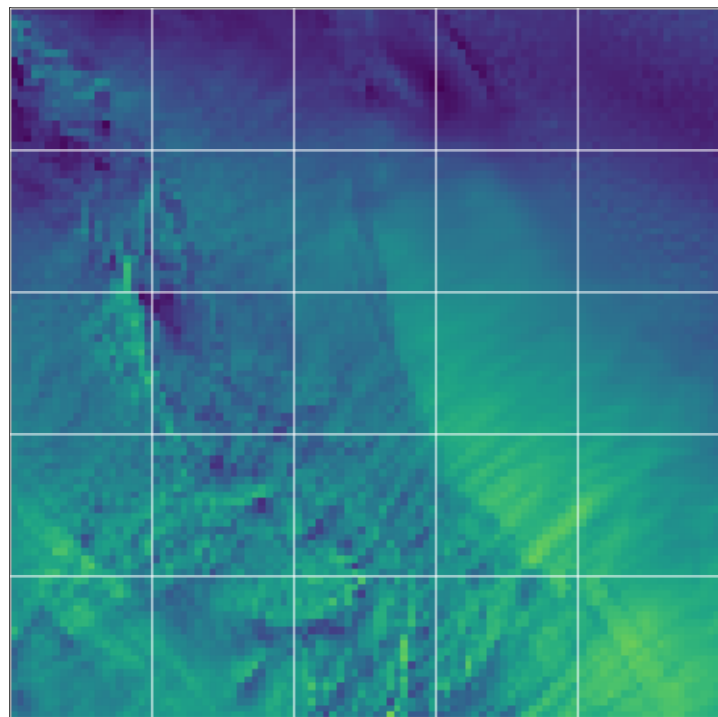
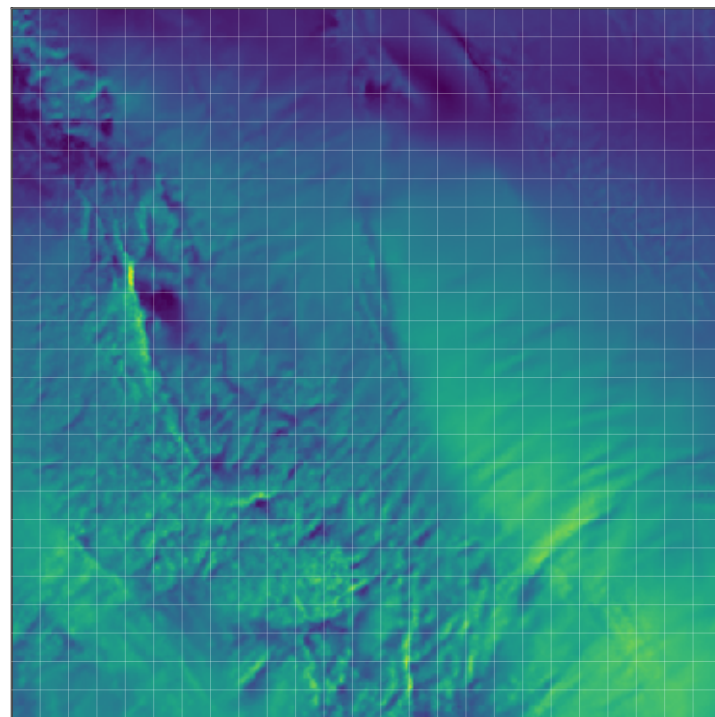


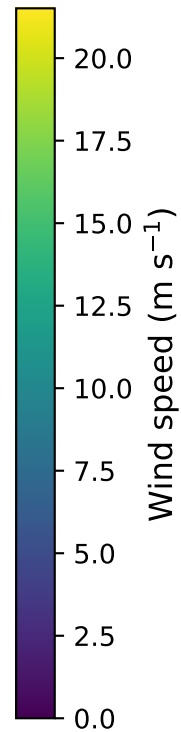
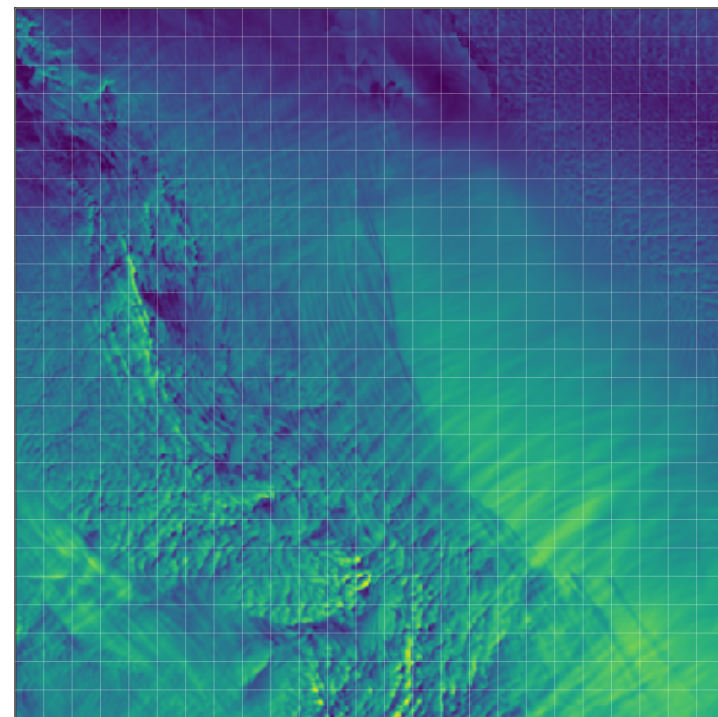
Wind-field super-resolution across the same physical region



SR model



compare



Lower-resolution input

$100 \times 100 \cdot 10 \text{ km / cell}$

SR prediction

$500 \times 500 \cdot 2 \text{ km / cell}$

High-resolution ground truth

$500 \times 500 \cdot 2 \text{ km / cell}$

SR predicts wind velocity (u, v) ; topology is evaluated on wind-speed magnitude $s = \sqrt{u^2 + v^2}$.

Topology metrics use the same fixed 160×160 SR/GT evaluation crop for every method.